

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		(remains) marine life / animals / plants (1) over millions of years / because of pressure / decompose in absence of oxygen (1)	2			2		
		(ii)		will run out / won't last forever / finite / cannot be replaced cannot be used again – neutral	1			1		
	(b)	(i)		distillation (1) accept any of following for (1) • boiling point • boiling temperature • sized molecules / chain length • different forces between molecules different temperature / different molecules / different melting point – neutral	2			2		
		(ii)		<u>mixture</u> element compound	1			1		
	(c)			84 / 83.7 / 83.72 (2) award (1) for 83 award (1) for M_r 86 allow ecf from incorrect M_r		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		B		1				
		(ii)		A C D – all three needed (1) B E – both needed (1) award (1) for one correct structure given in both parts		2		2	1	
	(e)			linked method and problem required burning / incineration – releases poisonous gases / toxic fumes / CO ₂ which causes global warming (1) send to landfill / bury – space running out / does not decompose / non-biodegradable (1) award (1) for two correct methods if insufficient descriptions of the problems	2			2		
				Question 4 total	8	5	0	13	3	0